Number Codes (Set B)

Year 4 Pt 2 · **Homework Worksheet**

Worked Example

Example: A code doubles a number and then adds 2. What is the code for 9?

Answer: 20 — 9 × 2 = 18, then 18 + 2 = 20.

Purple — Accessible (15 questions)

1. A diary lock uses this code: 1 = B, 4 = E, 9 = S. What is the code for the number 149?
2. A board game spinner uses this code: 2 = R, 3 = U, 5 = N. The code RUN stands for which number?
3. A bike chain lock uses this code: 1 = S, 4 = U, 6 = N. What is the code for 146? A) NUS B) USN C) SUN D) UNS E) NSU
4. An arcade token machine uses this code: 0 = F, 3 = A, 7 = R. What is the code for the number 730?
5. A museum keypad uses this code: 1 = D, 4 = O, 8 = G. The code OGD stands for which number?
6. A shed padlock uses this code: 2 = C, 3 = A, 7 = R. The code CAR stands for which number? A) 237 B) 273 C) 327 D) 372 E) 723
7. A campsite gate uses this code: 2 = T, 5 = E, 6 = N. What is the code for the number 265?
8. A science lab door uses this code: 1 = W, 3 = I, 7 = N. The code WIN stands for which number?
9. Using the code 0 = J, 1 = T, 2 = O, 3 = P, 4 = N, 5 = W, 6 = F, 7 = S, 8 = L, 9 = M, select ALL the numbers below whose code would END in the letter 'T'. A) 341 B) 472 C) 918 D) 251 E) 501
10. A tent zip uses this code: 1 = T, 4 = O, 9 = P. What is the code for 491? A) TOP B) POT C) TPO D) OTP E) OPT
11. A football scoreboard uses this code: 0 = G, 4 = O, 8 = L. What is the code for the number 408?
12. A train platform display uses this code: 2 = F, 3 = A, 5 = R. The code AFR stands for which number?
13. A diary padlock uses this code: 2 = O, 5 = L, 6 = D. The code OLD stands for which number? A) 265 B) 256 C) 526 D) 625 E) 652
14. A library card uses this code: 1 = C, 5 = A, 6 = T. What is the code for the number 516?
15. A space rocket control panel uses this code: 2 = M, 6 = O, 7 = N. The code MON stands for which number?

Green — Standard (15 questions)

1. A code is formed by doubling a number and then adding 4. What is the code for 5?
2. A code is formed by multiplying a number by 3 and then adding 1. The code for a number is 25. What was the original number?
3. A code is formed by doubling a number and then adding 7. What is the code for 6? A) 17 B) 18 C) 20 D) 19 E) 16
4. A code is formed by adding 2 to a number and then multiplying the result by 3. What is the code for 4?
5. A code is formed by halving a number and then adding 3. The code for a number is 11. What was the original number?
6. A code is formed by multiplying a number by 4 and then subtracting 3. The code for a number is 25. What was the original number? A) 6 B) 8 C) 7 D) 9 E) 5

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7. In a code, 2 → 9 and 4 → 17. Using the same rule, what is the code for 6?

8. In a code, 3 → 10 and 6 → 19. If the code for a number is 16, what is that number, using the same rule?

9. A code is formed by doubling a number and then adding 3. Select ALL the numbers below whose code would be LESS than 20. A) 6 B) 7 C) 8 D) 9 E) 10

10. In a code, 3 → 13 and 6 → 25. Using the same rule, what is the code for 8? A) 33 B) 32 C) 34 D) 31 E) 35

11. A code is formed by subtracting 1 from a number and then multiplying by 5. What is the code for 5?

12. A code is formed by multiplying a number by 2 and then subtracting 5. The code for a number is 9. What was the original number?

13. A code is formed by multiplying a number by 3 and then subtracting 4. What is the code for 7? A) 16 B) 17 C) 18 D) 19 E) 15

14. In a code, 4 → 14 and 7 → 23. Using the same rule, what is the code for 9?

15. In a code, 2 → 7 and 5 → 16. If the code for a number is 22, what is that number, using the same rule?

Standard / Advanced (15 questions)

1. A code doubles a number. What is the code of the code for 2? (Apply the doubling rule twice.)

2. In a code, every digit has 3 added to it, wrapping round after 9. What is the code of the code for 14? (Apply the rule twice.)

3. A code adds 6 to a number. The code for a certain number is 20. What is double that original number?

4. A code reverses the digits of a number and then adds 1. A certain number has the code 52. What do you get if you just reverse the digits of that original number, without adding 1?

5. A code doubles a number and then subtracts 1. The code for a number is 15. What is half of that original number?

6. In a code, the digits of a number are first reversed, and then each digit of the result has 2 added to it, wrapping round after 9. What is the code for 391? A) 315 B) 351 C) 135 D) 513 E) 153

7. Using the same two-step code (reverse the digits, then add 2 to each digit, wrapping after 9), the code for a number is 315. What was the original number?

8. A code first doubles a number, then reverses the digits of the result. What is the code for 23? A) 46 B) 24 C) 64 D) 62 E) 36

9. Using the same code (double the number, then reverse the digits of the result), the code for a number is 84. What was the original number?

10. A code triples a number and then adds together the digits of that result. What is the code for 24? A) 6 B) 9 C) 12 D) 72 E) 15

11. A code subtracts a number from 100, then reverses the digits of the result. What is the code for 37?

12. Using the same code (subtract the number from 100, then reverse the digits of the result), the code for a number is 21. What was the original number? A) 12 B) 79 C) 9 D) 88 E) 91

13. In a code, 12 → 25, 20 → 41, and 7 → 15. The same rule applies throughout. What is the code for 30?

14. A code doubles a number and then reverses the digits of the result. Select ALL the numbers below for which the final code would be GREATER than 50. A) 12 B) 23 C) 31 D) 8 E) 45

15. A code works in two stages: first subtract 7 from the number, then multiply the result by 2. If applying BOTH stages to a number gives a final result of 22, what was the original number?

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